

Predicting Sleep Quality from Daytime Wearable Metrics: An N-of-1 Machine Learning Study

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Introduction

Sleep quality is fundamental to cognitive function and metabolic health. Consumer wearables offer continuous physiological monitoring as an alternative to PSG.

Dataset: 158 consecutive nights (Nov 2025 – May 2026), Didiconn wrist-worn device.

Features (13): HR, HRV, SpO₂, steps, calories.

Task A – **Regression:** total sleep, REM, deep sleep (minutes).

Task B – **Classification:** poor sleep night ($\leq 82.25\%$, data-driven 25th percentile).

Research questions:

- (1) Can next-night sleep be predicted from daytime wearable data?
- (2) Which physiological indicators are early predictors of poor sleep?
- (3) Does low daytime HRV precede fragmented sleep?

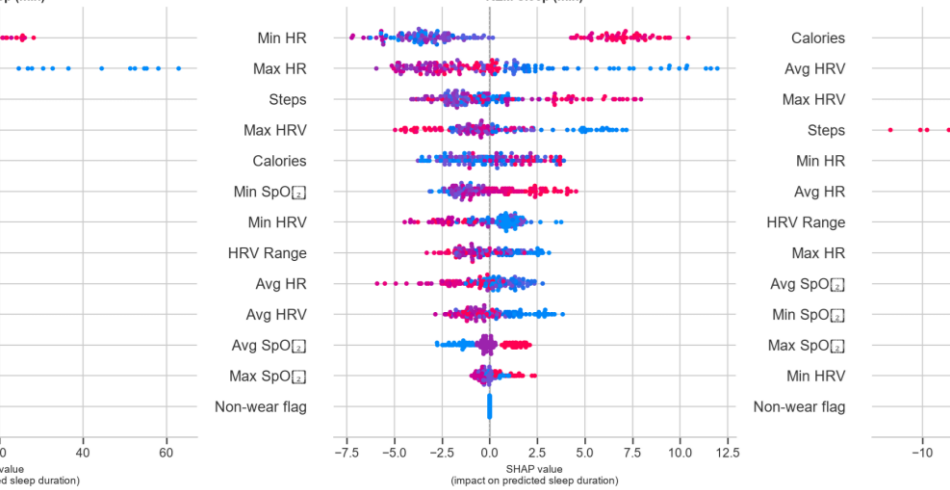


Fig. 1: SHAP beeswarm — feature importance for Total Sleep, REM, Deep Sleep. Calories and HRV dominate. Red = high feature value, blue = low.

Method

Preprocessing: t-1 lag alignment, nap deduplication, HRV range feature derived.

Validation: Walk-forward CV (TimeSeriesSplit, k=5, ~26 test nights/fold).

Random k-fold excluded — temporal leakage prevention.

ML models: Linear Regression, Ridge, Lasso, Random Forest, XGBoost (regression); Logistic Regression, SVM (RBF), RF, XGBoost (classification). class_weight=balanced.

DL models: LSTM + 1D-CNN, 7-day sliding window (N=151 sequences), Adam lr=1e-3.

Additional experiments (6): rolling features (49-dim), SOL/WASO targets, weekly patterns (Kruskal-Wallis), nested GridSearchCV tuning, multi-output regression, anomaly detection (GMM + Isolation Forest).

XAI: TreeSHAP — global importance, HRV dependence plots, waterfall explanations.

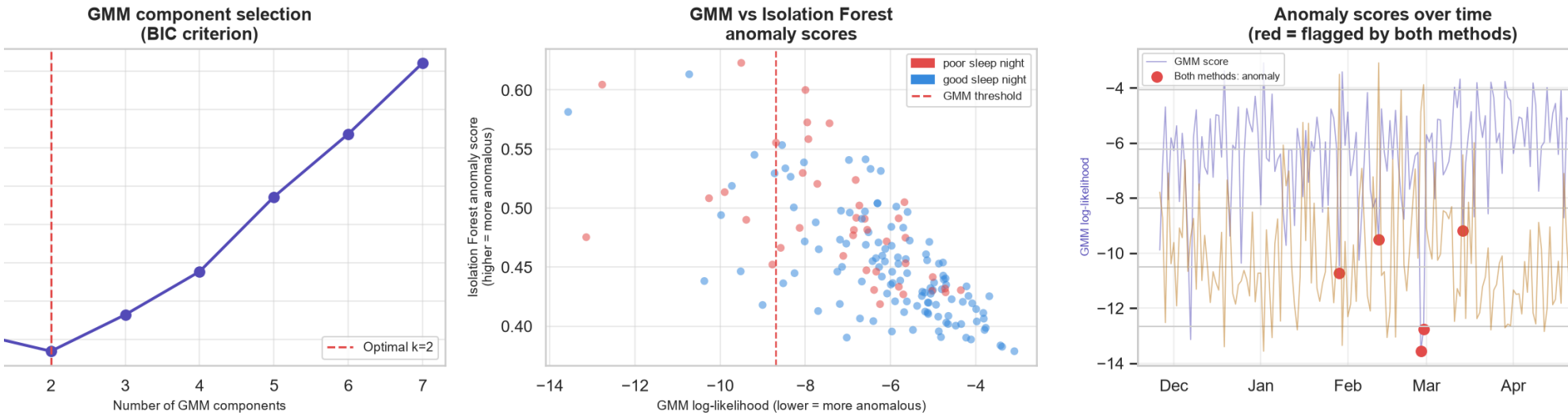


Fig. 2: Anomaly detection. Left: BIC curve (k=2 optimal). Centre: GMM vs IF scores by poor/good sleep label. Right: scores over 6 months.

Results

Regression — all models $R^2 < 0$ (worse than mean-prediction baseline):

- Random Forest: deep sleep $R^2 = -0.138$, total sleep $R^2 = -0.222$
 - XGBoost (tuned): total sleep $R^2 = -0.179$ ($\Delta +0.177$ vs default)
 - RF + rolling features: deep sleep $R^2 = -0.043$ ($\Delta +0.095$) ← largest gain
- Classification** — best supervised: SVM RBF AUC=0.622. F1 low (0.10–0.22). DL underperformed: LSTM AUC=0.446 (151 sequences insufficient).

Anomaly detection outperformed ALL supervised models:

- Isolation Forest AUC=0.728 | GMM AUC=0.696
 - 86.1% agreement; 5 nights flagged by both simultaneously
- SHAP:** Calories > Max HR > Max HRV > Steps.
HRV range = #1 feature for poor-sleep classification ($|SHAP|=0.042$).
Higher daytime HRV → less predicted sleep ($r \approx -0.75$) — post-exercise confound.

Weekly: sleep efficiency significant day-of-week effect (KW H=13.94, p=0.030).

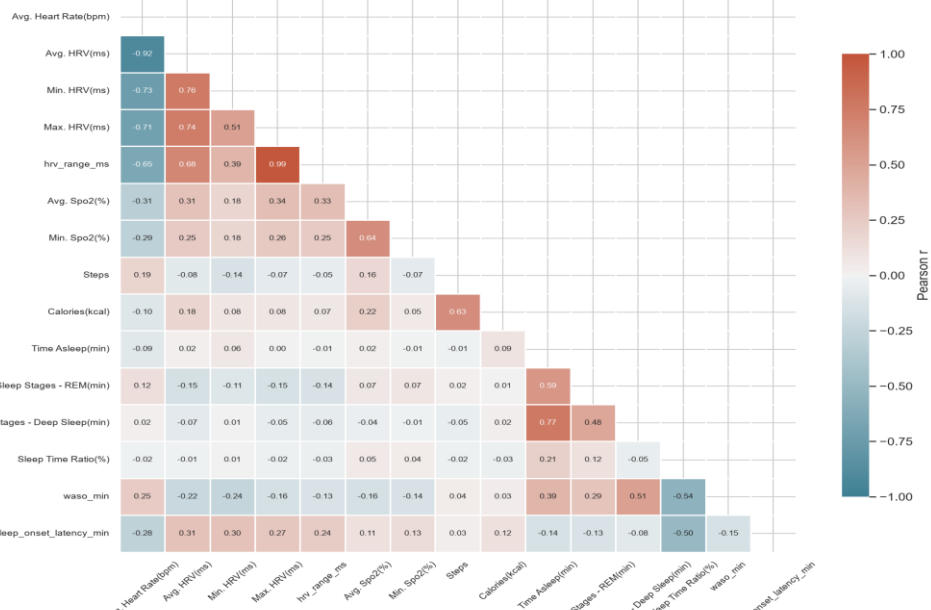


Fig. 3: Pearson correlation heatmap. Max $|r|=0.151$ (HRV vs REM, $p=0.059$) — weak linear associations between daytime physiology and sleep outcomes.

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Model Performance Summary

Deep Sleep: RF+Rolling $R^2 = -0.043$ (best)
Total Sleep: XGBoost (tuned) $R^2 = -0.179$
REM: Random Forest $R^2 = -0.275$
Classification: SVM AUC=0.622
Anomaly: IF AUC=0.728 ← best overall

Weekly patterns: Sleep efficiency showed significant day-of-week variation (Kruskal-Wallis H=13.94, p=0.030). Wednesday consistently worst.

DL underperformed: 151 sequences insufficient for LSTM/CNN to learn temporal dependencies.

Conclusions

Single-day wearable metrics show limited predictive power for sleep (all $R^2 < 0$), consistent with the multifactorial nature of sleep regulation.

Unsupervised anomaly detection (IF AUC=0.728) outperformed all supervised classifiers (best AUC=0.622) — poor sleep nights are structurally distinguishable in architecture.

Physiological trends > single values: rolling features gave largest R^2 gain (+0.095).

HRV variability (range), not average HRV, is the strongest poor-sleep classifier feature.

Limitations: N-of-1 (single participant), device not PSG-validated, daily HRV averages conflate resting/exercise/stress.

Future work: longer observation, continuous pre-sleep HRV, ecological momentary assessment.